

Discrepancy of Pseudorandom Vectors Generated by Elliptic Curves in the Non-Maximal-Period Regime

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Abstract

This paper studies Mok's open problem on discrepancy estimates for elliptic curve congruential generators in the regime $e \neq 1$ and non-maximal period $t \neq N = \#E(\mathbb{F}_q)$. Two families of results are obtained. First, in a full-coset regime characterized by a *relative maximal period condition* (RMPC) on an induced one-dimensional linear congruential generator, one proves Mok-type bounds of size $q^{1/2}/t$ for the extreme discrepancy D , the serial discrepancy D_s , and, under the corresponding derived RMPC, the non-overlapping discrepancy $\tilde{D}_s$. Second, in the general subperiod regime, one reduces D , D_s , and $\tilde{D}_s$ to Fourier ℓ^1 masses of admissible index sets attached to one-dimensional linear congruential generators. This yields unconditional baseline bounds of size $q^{1/2}/\sqrt{N^*}$, $q^{1/2}/\sqrt{N_s^*}$, and $q^{1/2}/\sqrt{\tilde{N}_s^*}$, respectively, and isolates the arithmetic bottleneck for further improvement.

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1 Introduction and Mok's problem in the regime $e \neq 1$

Let $E/\mathbb{F}_q$ be an elliptic curve with additive group law and identity element O . Fix an integer $e \neq 0$ and a point $Q \in E(\mathbb{F}_q)$. Consider the affine iteration

$$P_{n+1} = [e]P_n + Q, \quad n \geq 0, \quad (1)$$

starting from a point $P_0 \in E(\mathbb{F}_q)$. Throughout, P_0 is assumed to lie on a purely periodic orbit, and t denotes the least positive integer such that $P_t = P_0$.

Mok [1] established explicit discrepancy and serial discrepancy bounds in the translation case $e = 1$ by reducing Walsh coefficients to additive character sums over cyclic subgroups and then applying square-root cancellation for subgroup sums on elliptic curves. He also observed that if $e \neq 1$ but the period is maximal, $t = N = \#E(\mathbb{F}_q)$, then the same discrepancy estimate remains valid by reducing to a maximal translation orbit. The remaining problem is to estimate the extreme discrepancy D , the serial discrepancy D_s , and the non-overlapping discrepancy $\tilde{D}_s$ when $e \neq 1$ and $t \neq N$.

The main results proved below are of two kinds.

1. In a *full-coset regime* (Case I), the orbit does not necessarily cover $E(\mathbb{F}_q)$, but it covers a full coset of a cyclic subgroup canonically attached to (e, P_0, Q) . In this regime one recovers Mok-type bounds of size $q^{1/2}/t$.
2. In the *general subperiod regime* (Case II), the orbit is a proper subset of that coset. In this regime the discrepancy problem is reduced to the Fourier ℓ^1 mass of an admissible index set associated with a one-dimensional linear congruential generator.

2 Output map, discrepancy, Walsh coefficients, and subgroup sums

2.1 A trace-based digital output map

Write $q = p^{ar}$ with integers $a, r \geq 1$, and fix the tower $\mathbb{F}_q/\mathbb{F}_{p^a}/\mathbb{F}_p$. Choose an $\mathbb{F}_p$ -basis $\{\kappa_1, \dots, \kappa_a\}$ of $\mathbb{F}_{p^a}$ and an $\mathbb{F}_{p^a}$ -basis $\{\lambda_1, \dots, \lambda_r\}$ of $\mathbb{F}_q$. Following Wang [2], define for $\eta \in \mathbb{F}_q$ and $1 \leq j \leq r$,

$$\phi_j(\eta) := \sum_{i=1}^a \frac{\text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(\eta \lambda_j \kappa_i)}{p^i} \in [0, 1).$$

For $P \in E(\mathbb{F}_q) \setminus \{O\}$, define

$$\Phi(P) := (\phi_1(x(P)), \dots, \phi_r(x(P)), \phi_1(y(P)), \dots, \phi_r(y(P))) \in [0, 1)^{2r}. \quad (2)$$

This trace-based digitization is equivalent in spirit to Mok's map G : both maps are built from the trace coordinates of $x(P)$ and $y(P)$.

2.2 Point collections and discrepancy

Discrepancy is defined for finite *collections* of points, so multiplicities are counted. Let

$$\mathcal{P} = (v_0, \dots, v_{N-1}), \quad v_n \in [0, 1)^d.$$

For an axis-parallel box

$$B = \prod_{j=1}^d [u_j, v_j) \subset [0, 1)^d,$$

define

$$A(B; \mathcal{P}) := \#\{0 \leq n < N : v_n \in B\}.$$

The *extreme discrepancy* of $\mathcal{P}$ is

$$D(\mathcal{P}) := \sup_B \left| \frac{A(B; \mathcal{P})}{N} - \text{vol}(B) \right|,$$

where the supremum runs over all axis-parallel boxes in $[0, 1)^d$.

If $(v_n)_{n \geq 0}$ is periodic of period N , then for $2 \leq s \leq N$ the *serial s -tuple collection* is

$$\mathcal{P}^{(s)} := ((v_n, \dots, v_{n+s-1}))_{0 \leq n \leq N-1} \subset [0, 1)^{ds},$$

where the indices are taken modulo N . The serial discrepancy D_s is the extreme discrepancy of the admissible subcollection obtained after removing those blocks that hit the boundary (in the present setting, boundary points occur precisely when one of the underlying curve points equals O).

Following Mok [1], the *non-overlapping s -tuple collection* is

$$\tilde{\mathcal{P}}^{(s)} := ((v_{ns}, v_{ns+1}, \dots, v_{ns+s-1}))_{0 \leq n < N/\gcd(N, s)},$$

again with indices taken modulo N , and again after removing boundary blocks. Its discrepancy is denoted by $\tilde{D}_s$.

Throughout the paper, any discrepancy statement involving a quantity of the form

$$N^*, \quad N_s^*, \quad \tilde{N}_s^*$$

is understood under the natural assumption that the corresponding admissible point collection is nonempty.

2.3 Walsh discrepancy inequality

We use the base- p Walsh system. The only property needed here is a Hellekalek-type discrepancy inequality.

Lemma 1 (Walsh discrepancy inequality). *Let $\mathcal{P} = (v_0, \dots, v_{N-1}) \subset [0, 1]^d$ be a finite point collection lying on the grid $\{0, 1/p^a, \dots, (p^a - 1)/p^a\}^d$. Assume that for every nonzero Walsh frequency k in the relevant truncated set Δ_d^* one has*

$$|S(w_k, \mathcal{P})| \leq B,$$

where $S(w_k, \mathcal{P})$ denotes the normalized Walsh coefficient of w_k over $\mathcal{P}$. Then

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^d + B (C_0 \ln(p^a) + 1)^d, \quad (3)$$

where C_0 is an absolute constant (for example $C_0 = 1.78$ in Wang [2, Lemma 2.1]; Mok [1] uses a slightly larger explicit constant).

2.4 Walsh values as additive characters on $E(\mathbb{F}_q)$

Let

$$\psi_1(z) := \exp\left(\frac{2\pi i}{p} \text{Tr}_{\mathbb{F}_q/\mathbb{F}_p}(z)\right), \quad z \in \mathbb{F}_q,$$

be the canonical additive character of $\mathbb{F}_q$.

Lemma 2 (Walsh-to-character conversion). *For each nonzero Walsh frequency $k \in \Delta_{2r}^*$ there exist $\eta_k, \tilde{\eta}_k \in \mathbb{F}_q$, not both zero, such that for all $P \in E(\mathbb{F}_q) \setminus \{O\}$,*

$$w_k(\Phi(P)) = \psi_1(\eta_k x(P) + \tilde{\eta}_k y(P)).$$

Consequently, for any finite $S \subset E(\mathbb{F}_q)$ and the point collection

$$\mathcal{P} = (\Phi(P))_{P \in S, P \neq O},$$

one has

$$|\mathcal{P}| \cdot S(w_k, \mathcal{P}) = \sum_{P \in S, P \neq O} \psi_1(\eta_k x(P) + \tilde{\eta}_k y(P)), \quad (4)$$

where the sum on the right counts P with the same multiplicity as in $\mathcal{P}$.

Proof. This is the standard computation for the trace-based digit map; see Wang [2, (2.2)–(2.3)]. $\square$

2.5 Twisted subgroup character sums

The following square-root cancellation estimate is the main analytic input.

Lemma 3 (Kohel–Shparlinski bound). *Let $H \subset E(\mathbb{F}_q)$ be a subgroup. Let $f \in \mathbb{F}_q(E)$ be a rational function that is not of Artin–Schreier form $g^p - g$. Let ψ be a nontrivial additive character of $\mathbb{F}_q$, and let ω be a group character of H . Then*

$$\left| \sum_{\substack{P \in H \\ f(P) \neq \infty}} \omega(P) \psi(f(P)) \right| \leq 2 \deg(f) q^{1/2}. \quad (5)$$

Moreover, if the polar divisor of f is supported on a single place, then

$$\left| \sum_{\substack{P \in H \\ f(P) \neq \infty}} \omega(P) \psi(f(P)) \right| \leq (1 + \deg(f)) q^{1/2}. \quad (6)$$

2.6 Auxiliary facts on separable multiplication maps and Artin–Schreier poles

Two standard facts are used repeatedly in the proofs.

Lemma 4 (Pullback under separable multiplication maps). *Let $n \in \mathbb{Z}$ with $\gcd(n, p) = 1$. Then $[n] : E \rightarrow E$ is a separable isogeny of degree n^2 . Moreover:*

- (i) *for every $A \in E(\overline{\mathbb{F}}_q)$, the translation map $\tau_A(P) := P + A$ is an automorphism of degree 1;*
- (ii) *for every nonconstant $f \in \mathbb{F}_q(E)$,*

$$\deg(f \circ \tau_A) = \deg(f), \quad \deg(f \circ [n]) = n^2 \deg(f),$$

and hence

$$\deg(f \circ (\tau_A \circ [n])) = n^2 \deg(f);$$

- (iii) *if f has a pole of order m at $Q \in E(\overline{\mathbb{F}}_q)$, then $f \circ (\tau_A \circ [n])$ has poles of order m at the points of $[n]^{-1}(Q - A)$. In particular, since $[n]$ is separable, this pole set has cardinality n^2 over $\overline{\mathbb{F}}_q$.*

Proof. The map $[n]$ is a separable isogeny of degree n^2 whenever $\gcd(n, p) = 1$, and translations are automorphisms. The degree identities follow from standard facts on morphisms of curves and pullback of rational functions. For the pole statement, if f has a pole of order m at Q , then $f \circ \tau_A$ has a pole of order m at $Q - A$. Pulling back by the separable map $[n]$ preserves the pole order at each point of the fibre and yields exactly n^2 distinct preimages. $\square$

Lemma 5 (A pole-order criterion excluding Artin–Schreier form). *Let $f \in \mathbb{F}_q(E)$. If f has a pole whose order is not divisible by p , then f is not of the form $h^p - h$ with $h \in \mathbb{F}_q(E)$.*

Proof. If h has a pole of order m at some point of $E(\overline{\mathbb{F}}_q)$, then $h^p - h$ has a pole of order pm there, since the term h^p dominates h at the pole. Therefore every pole order of a function of the form $h^p - h$ is divisible by p . The claim follows by contraposition. $\square$

3 The full-coset regime (Case I)

Throughout this section assume $|e| \geq 2$ and $\gcd(e, p) = 1$.

3.1 Orbit reduction to a cyclic coset

Lemma 6 (Orbit reduction to a cyclic coset). *Let (P_n) satisfy (1). Define*

$$\beta_n := \frac{e^n - 1}{e - 1} \in \mathbb{Z} \quad (n \geq 0, \beta_0 = 0), \quad R := P_1 - P_0 = (e - 1)P_0 + Q \in E(\mathbb{F}_q).$$

Then for all $n \geq 0$,

$$P_n = P_0 + [\beta_n]R. \quad (7)$$

In particular, the full orbit is contained in the coset $P_0 + \langle R \rangle$.

Proof. By induction on n one first obtains

$$P_n = [e^n]P_0 + [\beta_n]Q,$$

using the recursion $\beta_{n+1} = e\beta_n + 1$. Subtracting P_0 gives

$$P_n - P_0 = [e^n - 1]P_0 + [\beta_n]Q = [\beta_n]((e - 1)P_0 + Q) = [\beta_n]R,$$

which is (7). □

Lemma 7 (Pure periodicity forces $\gcd(e, m) = 1$). *With $m = \text{ord}(R)$ as above, one has*

$$\gcd(e, m) = 1.$$

Proof. Let $d := \gcd(e, m)$. If $d > 1$, then the defining recursion $\beta_{n+1} = e\beta_n + 1$ gives, modulo d ,

$$\beta_{n+1} \equiv e\beta_n + 1 \equiv 1 \pmod{d} \quad (n \geq 0),$$

because $d \mid e$. Hence

$$\beta_n \equiv 1 \pmod{d} \quad \text{for all } n \geq 1,$$

while $\beta_0 \equiv 0 \pmod{d}$. On the other hand, periodicity of the orbit implies $P_t = P_0$, so Lemma 6 gives

$$[\beta_t]R = O.$$

Since $\text{ord}(R) = m$, it follows that $m \mid \beta_t$, hence $d \mid \beta_t$. This contradicts $\beta_t \equiv 1 \pmod{d}$. Therefore $d = 1$. □

Let

$$m := \text{ord}(R), \quad H := \langle R \rangle \subset E(\mathbb{F}_q).$$

Reducing (7) modulo H , one sees that the orbit period equals the period of the index sequence (β_n) modulo m . Moreover,

$$\beta_{n+1} \equiv e\beta_n + 1 \pmod{m}, \quad \beta_0 \equiv 0. \tag{8}$$

Thus the affine generator on $E(\mathbb{F}_q)$ induces a one-dimensional LCG on $\mathbb{Z}/m\mathbb{Z}$.

3.2 The relative maximal period condition

Definition 8 (RMPC). The triple (e, P_0, Q) is said to satisfy the *relative maximal period condition* if the LCG (8) has full period $m = \text{ord}(R)$, that is,

$$\{\beta_0, \dots, \beta_{m-1}\} = \mathbb{Z}/m\mathbb{Z}.$$

Equivalently, the orbit point set is the full coset

$$\{P_0, \dots, P_{m-1}\} = P_0 + \langle R \rangle.$$

Lemma 9 (Hull–Dobell criterion for RMPC). *The LCG (8) has full period m if and only if:*

(i) *for every prime $\ell \mid m$, one has $\ell \mid (e - 1)$;*

(ii) *if $4 \mid m$, then $4 \mid (e - 1)$.*

Proof. This is the classical Hull–Dobell criterion for the maximal period of the LCG $x_{n+1} \equiv ex_n + 1 \pmod{m}$. □

3.3 Extreme discrepancy under RMPC

Theorem 10 (Extreme discrepancy in the full-coset regime). *Assume RMPC and let $t = m = \text{ord}(R)$. Define the admissible point collection*

$$\mathcal{P} := (\Phi(P_n))_{\substack{0 \leq n \leq t-1 \\ P_n \neq O}}, \quad N^* := |\mathcal{P}|.$$

Assume $p \geq 5$. Then

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} (C_0 \ln(p^a) + 1)^{2r}. \quad (9)$$

Proof. Under RMPC, Lemma 6 gives the set identity

$$\{P_0, \dots, P_{t-1}\} = \{P_0 + [u]R : u \in \mathbb{Z}/t\mathbb{Z}\}.$$

Since discrepancy is invariant under permutation of the underlying point collection, we may re-index by $u \in \{0, \dots, t-1\}$.

Fix a nonzero Walsh frequency $k \in \Delta_{2r}^*$. By Lemma 2,

$$N^* S(w_k, \mathcal{P}) = \sum_{\substack{u=0 \\ P_0 + [u]R \neq O}}^{t-1} \psi_1(\eta_k x(P_0 + [u]R) + \tilde{\eta}_k y(P_0 + [u]R)).$$

Define

$$f_{k,P_0}(P) := \eta_k x(P + P_0) + \tilde{\eta}_k y(P + P_0) \in \mathbb{F}_q(E).$$

The polar divisor of f_{k,P_0} is supported at the single point $-P_0$, and $\deg(f_{k,P_0}) \leq 3$. Since $p \geq 5$, its pole order is 2 or 3, hence not divisible by p ; therefore Lemma 5 shows that f_{k,P_0} is not of Artin–Schreier form. Applying Lemma 3 in the single-pole form (6) to $H = \langle R \rangle$ yields

$$\left| \sum_{\substack{P \in H \\ f_{k,P_0}(P) \neq \infty}} \psi_1(f_{k,P_0}(P)) \right| \leq (1 + \deg(f_{k,P_0})) q^{1/2} \leq 4q^{1/2}.$$

Hence $|S(w_k, \mathcal{P})| \leq 4q^{1/2}/N^*$ for every nonzero Walsh frequency. Applying Lemma 1 with $d = 2r$ proves (9). $\square$

3.4 Serial discrepancy under RMPC

Theorem 11 (Serial discrepancy in the full-coset regime). *Assume RMPC, $|e| \geq 2$, $\gcd(e, p) = 1$, and $p \geq 5$. Fix $2 \leq s < t$. Let $\mathcal{P}^{(s)}$ be the admissible serial s -tuple collection attached to the orbit (P_n) , and let $N_s^* := |\mathcal{P}^{(s)}|$. Then*

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2} \left(\sum_{j=0}^{s-1} e^{2j}\right) + s}{N_s^*} (C_0 \ln(p^a) + 1)^{2rs}. \quad (10)$$

Proof. Under RMPC, the admissible starting points may be parameterized by $u \in \mathbb{Z}/t\mathbb{Z}$ with $t = \text{ord}(R)$. Using the identity

$$\beta_{n+j} = \beta_j + e^j \beta_n$$

in $\mathbb{Z}$, one obtains

$$P_{n+j} = A_j + [e^j u]R, \quad A_j := P_0 + [\beta_j]R, \quad u := \beta_n \bmod t.$$

For each $0 \leq j \leq s-1$, define the exceptional set

$$\mathcal{E}_j := \{u \in \mathbb{Z}/t\mathbb{Z} : A_j + [e^j u]R = O\}.$$

RMPC implies that every prime divisor of t divides $e-1$, hence $\gcd(e, t) = 1$. Therefore multiplication by e^j on $\mathbb{Z}/t\mathbb{Z}$ is bijective. The equation

$$A_j + [e^j u]R = O$$

is equivalent to

$$[e^j u]R = -A_j,$$

and thus has either no solution or exactly one solution in $\mathbb{Z}/t\mathbb{Z}$. Hence

$$|\mathcal{E}_j| \leq 1 \quad (0 \leq j \leq s-1),$$

so the admissible index set

$$U_s := \{u \in \mathbb{Z}/t\mathbb{Z} : A_j + [e^j u]R \neq O \text{ for all } 0 \leq j \leq s-1\}$$

satisfies

$$|\mathbb{Z}/t\mathbb{Z} \setminus U_s| \leq s. \quad (11)$$

The serial point collection $\mathcal{P}^{(s)}$ is the image of U_s under

$$u \longmapsto (\Phi(A_0 + [u]R), \Phi(A_1 + [eu]R), \dots, \Phi(A_{s-1} + [e^{s-1}u]R)).$$

Fix a nonzero Walsh frequency $k \in \Delta_{2rs}^*$. Applying Lemma 2 coordinatewise, there exist coefficients $(\eta_{k,j}, \tilde{\eta}_{k,j}) \in \mathbb{F}_q^2$, not all zero, such that

$$N_s^* S(w_k, \mathcal{P}^{(s)}) = \sum_{u \in U_s} \psi_1(g_k([u]R)),$$

where

$$g_k(P) := \sum_{j=0}^{s-1} \left(\eta_{k,j} x([e^j]P + A_j) + \tilde{\eta}_{k,j} y([e^j]P + A_j) \right) \in \mathbb{F}_q(E).$$

Degree bound. By Lemma 4, since $\gcd(e, p) = 1$, each map $[e^j]$ is separable of degree e^{2j} and translation preserves degree. Hence

$$\deg(x([e^j]P + A_j)) = 2e^{2j}, \quad \deg(y([e^j]P + A_j)) = 3e^{2j},$$

so

$$\deg(g_k) \leq 3 \sum_{j=0}^{s-1} e^{2j}. \quad (12)$$

A pole of order 2 or 3. Let j_0 be maximal with $(\eta_{k,j_0}, \tilde{\eta}_{k,j_0}) \neq (0, 0)$. By Lemma 4, the j_0 -th summand has pole set $[e^{j_0}]^{-1}(-A_{j_0})$, of cardinality e^{2j_0} over $\mathbb{F}_q$, and each such pole has order 2 or

3. For each $j < j_0$, the j -th summand has at most e^{2j} poles. Hence the union of all pole sets for $j < j_0$ has cardinality at most

$$\sum_{j=0}^{j_0-1} e^{2j} < e^{2j_0} \quad (|e| \geq 2).$$

Therefore there exists a point P^* which is a pole of the j_0 -th summand and not a pole of any smaller- j summand. At P^* the j_0 -th summand has pole order 2 or 3. The x - and y -contributions cannot cancel there because their pole orders are different. Thus g_k itself has a pole of order 2 or 3 at P^* . Since $p \geq 5$, this pole order is not divisible by p , and Lemma 5 implies that g_k is not of Artin–Schreier form.

Comparison with the full finite-value set. Let

$$U_k := \{u \in \mathbb{Z}/t\mathbb{Z} : g_k([u]R) \neq \infty\}.$$

Then $U_s \subseteq U_k$, because every admissible block makes each coordinate finite. Therefore

$$U_k \setminus U_s \subseteq \mathbb{Z}/t\mathbb{Z} \setminus U_s,$$

and (11) gives

$$|U_k \setminus U_s| \leq s.$$

Since each summand $\psi_1(g_k([u]R))$ has absolute value 1 on U_k , one obtains

$$\left| \sum_{u \in U_s} \psi_1(g_k([u]R)) \right| \leq \left| \sum_{u \in U_k} \psi_1(g_k([u]R)) \right| + s.$$

The first sum is a complete finite-value subgroup sum on $H = \langle R \rangle$, so Lemma 3 and (12) give

$$\left| \sum_{u \in U_k} \psi_1(g_k([u]R)) \right| \leq 2 \deg(g_k) q^{1/2} \leq 6q^{1/2} \sum_{j=0}^{s-1} e^{2j}.$$

Thus

$$\left| N_s^* S(w_k, \mathcal{P}^{(s)}) \right| \leq 6q^{1/2} \sum_{j=0}^{s-1} e^{2j} + s.$$

Applying Lemma 1 with $d = 2rs$ proves (10). $\square$

Remark 12. If none of the cosets $A_j + \langle R \rangle$ contains O for $0 \leq j \leq s-1$, then the boundary correction term $+s$ in (10) disappears. In particular, when the entire coset $P_0 + \langle R \rangle$ avoids O , one obtains the sharper bound with numerator $6q^{1/2} \sum_{j=0}^{s-1} e^{2j}$.

3.5 A period lemma for decimated subsequences

Lemma 13 (Period of the s -step subsequence). *Let (P_n) have period t , and fix $s \geq 1$. Then the decimated subsequence $(P_{ns})_{n \geq 0}$ has period*

$$L_s := \frac{t}{\gcd(t, s)}.$$

Proof. Since $L_s s$ is divisible by t , one has

$$P_{(n+L_s)s} = P_{ns+L_ss} = P_{ns}$$

for all n . Thus the period divides L_s . Conversely, if $r \geq 1$ satisfies $P_{(n+r)s} = P_{ns}$ for all n , then in particular $P_{rs} = P_0$. Since (P_n) has period t , it follows that $t \mid rs$. By minimality of $L_s = t/\gcd(t, s)$, one has $L_s \mid r$. Hence the period of (P_{ns}) is exactly L_s . $\square$

3.6 Non-overlapping discrepancy under a derived RMPC

Definition 14 (Derived RMPC). Fix $s \geq 2$ and define

$$Q_s := \sum_{j=0}^{s-1} [e^j]Q, \quad T_s(P) := T^s(P) = [e^s]P + Q_s, \quad R_s := T_s(P_0) - P_0 = P_s - P_0.$$

Let $m_s := \text{ord}(R_s)$. The *derived RMPC of step s* is the requirement that the LCG

$$\gamma_{n+1} \equiv e^s \gamma_n + 1 \pmod{m_s}, \quad \gamma_0 \equiv 0,$$

has full period m_s .

Theorem 15 (Non-overlapping discrepancy in the full-coset regime). *Assume $|e| \geq 2$, $\gcd(e, p) = 1$, and $p \geq 5$. Fix $s \geq 2$ and assume the derived RMPC of step s . Let $\tilde{\mathcal{P}}^{(s)}$ be the admissible non-overlapping s -tuple collection, and let $\tilde{N}_s^* := |\tilde{\mathcal{P}}^{(s)}|$. Then*

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2} \left(\sum_{j=0}^{s-1} e^{2j}\right) + s}{\tilde{N}_s^*} (C_0 \ln(p^a) + 1)^{2rs}. \quad (13)$$

Proof. The subsequence $(P_{ns})_{n \geq 0}$ satisfies

$$P_{(n+1)s} = T_s(P_{ns}) = [e^s]P_{ns} + Q_s.$$

Applying Lemma 6 to the derived system (T_s, P_0) gives

$$P_{ns} = P_0 + [\gamma_n]R_s, \quad \gamma_n = \frac{e^{sn} - 1}{e^s - 1}.$$

By Lemma 13, the subsequence (P_{ns}) has period

$$L_s = \frac{t}{\gcd(t, s)}.$$

Under derived RMPC, the sequence (γ_n) has full period m_s modulo m_s . Since $P_{ns} = P_0 + [\gamma_n]R_s$ and $\text{ord}(R_s) = m_s$, it follows that the subsequence (P_{ns}) traverses the full coset $P_0 + \langle R_s \rangle$ and in particular

$$L_s = m_s.$$

For $0 \leq j \leq s-1$ and $n \geq 0$,

$$P_{ns+j} = T^j(P_{ns}) = P_j + [e^j \gamma_n]R_s, \quad P_j := T^j(P_0).$$

Therefore the admissible non-overlapping s -blocks are parameterized by

$$u \in \mathbb{Z}/m_s\mathbb{Z}$$

via

$$u \longmapsto (\Phi(P_0 + [u]R_s), \Phi(P_1 + [eu]R_s), \dots, \Phi(P_{s-1} + [e^{s-1}u]R_s)).$$

For each $0 \leq j \leq s-1$, define

$$\tilde{\mathcal{E}}_j := \{u \in \mathbb{Z}/m_s\mathbb{Z} : P_j + [e^j u]R_s = O\}.$$

Derived RMPC implies that every prime divisor of m_s divides $e^s - 1$, hence $\gcd(e, m_s) = 1$. Therefore multiplication by e^j on $\mathbb{Z}/m_s\mathbb{Z}$ is bijective. The equation

$$P_j + [e^j u]R_s = O$$

is equivalent to

$$[e^j u]R_s = -P_j,$$

and hence has either no solution or exactly one solution in $\mathbb{Z}/m_s\mathbb{Z}$. Thus

$$|\tilde{\mathcal{E}}_j| \leq 1 \quad (0 \leq j \leq s-1),$$

so the admissible index set

$$\tilde{U}_s := \{u \in \mathbb{Z}/m_s\mathbb{Z} : P_j + [e^j u]R_s \neq O \text{ for all } 0 \leq j \leq s-1\}$$

satisfies

$$|\mathbb{Z}/m_s\mathbb{Z} \setminus \tilde{U}_s| \leq s.$$

Fix $k \in \Delta_{2rs}^* \setminus \{0\}$. Applying Lemma 2 coordinatewise, one obtains coefficients $(\eta_{k,j}, \tilde{\eta}_{k,j}) \in \mathbb{F}_q^2$, not all zero, such that

$$\tilde{N}_s^* S(w_k, \tilde{\mathcal{P}}^{(s)}) = \sum_{u \in \tilde{U}_s} \psi_1(\tilde{g}_k([u]R_s)),$$

where

$$\tilde{g}_k(P) := \sum_{j=0}^{s-1} \left(\eta_{k,j} x([e^j]P + P_j) + \tilde{\eta}_{k,j} y([e^j]P + P_j) \right) \in \mathbb{F}_q(E).$$

Exactly as in the proof of Theorem 11, using Lemma 4 and Lemma 5, one has

$$\deg(\tilde{g}_k) \leq 3 \sum_{j=0}^{s-1} e^{2j},$$

and $\tilde{g}_k$ is not of Artin–Schreier form. Let

$$\tilde{U}_k := \{u \in \mathbb{Z}/m_s\mathbb{Z} : \tilde{g}_k([u]R_s) \neq \infty\}.$$

Then $\tilde{U}_s \subseteq \tilde{U}_k$, and

$$\tilde{U}_k \setminus \tilde{U}_s \subseteq \mathbb{Z}/m_s\mathbb{Z} \setminus \tilde{U}_s.$$

Since $|\mathbb{Z}/m_s\mathbb{Z} \setminus \tilde{U}_s| \leq s$, it follows that

$$|\tilde{U}_k \setminus \tilde{U}_s| \leq s.$$

Therefore

$$\left| \sum_{u \in \tilde{U}_s} \psi_1(\tilde{g}_k([u]R_s)) \right| \leq \left| \sum_{u \in \tilde{U}_k} \psi_1(\tilde{g}_k([u]R_s)) \right| + s.$$

Applying Lemma 3 to the cyclic subgroup $\langle R_s \rangle$ yields

$$\left| \sum_{u \in \tilde{U}_k} \psi_1(\tilde{g}_k([u]R_s)) \right| \leq 2 \deg(\tilde{g}_k) q^{1/2} \leq 6q^{1/2} \sum_{j=0}^{s-1} e^{2j}.$$

Therefore

$$\left| \tilde{N}_s^* S(w_k, \tilde{\mathcal{P}}^{(s)}) \right| \leq 6q^{1/2} \sum_{j=0}^{s-1} e^{2j} + s.$$

Applying Lemma 1 in dimension $2rs$ proves (13). $\square$

Remark 16. If none of the cosets $P_j + \langle R_s \rangle$ contains O for $0 \leq j \leq s-1$, then the boundary correction term $+s$ in (13) disappears.

4 The general subperiod regime (Case II)

We retain the notation

$$R = (e-1)P_0 + Q, \quad m = \text{ord}(R), \quad H = \langle R \rangle,$$

and we write

$$B = \{\beta_0, \dots, \beta_{t-1}\} \subset \mathbb{Z}/m\mathbb{Z}$$

for the index cycle associated with the LCG (8). Since (P_n) has exact period t and $P_n = P_0 + [\beta_n]R$, the residues $\beta_0, \dots, \beta_{t-1}$ are pairwise distinct modulo m .

4.1 Completion on a cyclic subgroup at arbitrary modulus

Definition 17 (Fourier transform and Fourier ℓ^1 mass at modulus M). Let $M \geq 1$ and let $A \subset \mathbb{Z}/M\mathbb{Z}$. Define

$$\widehat{\mathbf{1}}_A^{(M)}(j) := \sum_{u \in A} e^{-2\pi i j u / M}, \quad 0 \leq j < M,$$

and

$$\mathcal{L}_M(A) := \frac{1}{M} \sum_{j=0}^{M-1} \left| \widehat{\mathbf{1}}_A^{(M)}(j) \right|.$$

Lemma 18 (Exact completion on a cyclic subgroup). *Let $S \in E(\mathbb{F}_q)$ have order M , and let $H_S = \langle S \rangle$. For every subset $A \subset \mathbb{Z}/M\mathbb{Z}$ and every function $F : H_S \rightarrow \mathbb{C}$,*

$$\sum_{u \in A} F([u]S) = \frac{1}{M} \sum_{j=0}^{M-1} \widehat{\mathbf{1}}_A^{(M)}(j) \sum_{P \in H_S} \omega_j(P) F(P), \quad (14)$$

where $\omega_j([u]S) := e^{2\pi i j u / M}$.

Proof. This is the orthogonality relation on $\mathbb{Z}/M\mathbb{Z}$:

$$\mathbf{1}_A(u) = \frac{1}{M} \sum_{j=0}^{M-1} \widehat{\mathbf{1}}_A^{(M)}(j) e^{2\pi i j u / M}.$$

Substituting into $\sum_u \mathbf{1}_A(u) F([u]S)$ yields (14). $\square$

Proposition 19 (Orbit-sum bound via Fourier ℓ^1 mass). *Let $S \in E(\mathbb{F}_q)$ have order M , let $A \subset \mathbb{Z}/M\mathbb{Z}$, and let $f \in \mathbb{F}_q(E)$ be a rational function that is not of Artin–Schreier form. Then*

$$\left| \sum_{\substack{u \in A \\ f([u]S) \neq \infty}} \psi_1(f([u]S)) \right| \leq 2 \deg(f) q^{1/2} \mathcal{L}_M(A). \quad (15)$$

If the polar divisor of f is supported on a single place, one may replace $2 \deg(f)$ by $1 + \deg(f)$.

Proof. Define

$$F(P) := \begin{cases} \psi_1(f(P)), & f(P) \neq \infty, \\ 0, & f(P) = \infty. \end{cases}$$

Applying Lemma 18 with this F , the inner sums become exactly the finite-value twisted subgroup sums appearing in Lemma 3. Taking absolute values and summing with weights $|\widehat{\mathbf{1}}_A^{(M)}(j)|/M$ yields (15). $\square$

4.2 Admissible index sets

For one-point discrepancy, define

$$B_{\text{adm}} := \{u \in B : P_0 + [u]R \neq O\}.$$

Then

$$|B_{\text{adm}}| = N^*, \quad |B \setminus B_{\text{adm}}| \leq 1.$$

For serial discrepancy of length s , define

$$B_{s,\text{adm}} := \{u \in B : P_0 + [\beta_j + e^j u]R \neq O \text{ for all } 0 \leq j \leq s-1\}.$$

Then

$$|B_{s,\text{adm}}| = N_s^*.$$

Moreover, for each fixed j , the map

$$u \mapsto \beta_j + e^j u \pmod{m}$$

coincides with the j -fold iterate of the LCG map $x \mapsto ex + 1$ on the cycle B , hence is a permutation of B . Therefore each coordinate condition excludes at most one element of B , so

$$|B \setminus B_{s,\text{adm}}| \leq s. \quad (16)$$

For the non-overlapping variant, fix $s \geq 2$ and keep the notation

$$Q_s := \sum_{j=0}^{s-1} [e^j]Q, \quad T_s(P) := T^s(P) = [e^s]P + Q_s, \quad R_s := T_s(P_0) - P_0.$$

Let

$$m_s := \text{ord}(R_s), \quad L_s := \frac{t}{\gcd(t, s)}.$$

Applying Lemma 6 to the derived system (T_s, P_0) gives

$$P_{ns} = P_0 + [\gamma_n]R_s, \quad \gamma_{n+1} \equiv e^s \gamma_n + 1 \pmod{m_s}, \quad \gamma_0 \equiv 0.$$

By Lemma 13, the subsequence (P_{ns}) has period L_s . Set

$$C_s := \{\gamma_0, \dots, \gamma_{L_s-1}\} \subset \mathbb{Z}/m_s\mathbb{Z}.$$

Since the decimated subsequence (P_{ns}) has exact period L_s and $P_{ns} = P_0 + [\gamma_n]R_s$, the residues $\gamma_0, \dots, \gamma_{L_s-1}$ are pairwise distinct modulo m_s . Then define the derived admissible index set

$$\tilde{B}_{s,\text{adm}} := \{u \in C_s : P_j + [e^j u]R_s \neq O \text{ for all } 0 \leq j \leq s-1\}.$$

By construction,

$$|\tilde{B}_{s,\text{adm}}| = \tilde{N}_s^*.$$

Lemma 20 (Removing a small set changes Fourier mass by at most its size). *Let $M \geq 1$ and let $A \subseteq B \subseteq \mathbb{Z}/M\mathbb{Z}$. Then*

$$\mathcal{L}_M(A) \leq \mathcal{L}_M(B) + |B \setminus A|.$$

In particular,

$$\mathcal{L}_m(B_{\text{adm}}) \leq \mathcal{L}_m(B) + 1, \quad \mathcal{L}_m(B_{s,\text{adm}}) \leq \mathcal{L}_m(B) + s.$$

Proof. Write $B = A \sqcup C$ with $C = B \setminus A$. Then

$$\widehat{\mathbf{1}}_A^{(M)}(j) = \widehat{\mathbf{1}}_B^{(M)}(j) - \widehat{\mathbf{1}}_C^{(M)}(j),$$

so

$$\left| \widehat{\mathbf{1}}_A^{(M)}(j) \right| \leq \left| \widehat{\mathbf{1}}_B^{(M)}(j) \right| + \left| \widehat{\mathbf{1}}_C^{(M)}(j) \right|.$$

Summing over j and dividing by M gives

$$\mathcal{L}_M(A) \leq \mathcal{L}_M(B) + \mathcal{L}_M(C).$$

Since

$$\sum_{j=0}^{M-1} \left| \widehat{\mathbf{1}}_C^{(M)}(j) \right| \leq M|C|$$

trivially, one has $\mathcal{L}_M(C) \leq |C|$. □

4.3 Extreme discrepancy in the general subperiod regime

Theorem 21 (General extreme discrepancy bound via admissible Fourier mass). *Let*

$$\mathcal{P} := (\Phi(P_n))_{0 \leq n \leq t-1, \atop P_n \neq O}, \quad N^* := |\mathcal{P}|.$$

Assume $p \geq 5$. Then

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} \mathcal{L}_m(B_{\text{adm}}) (C_0 \ln(p^a) + 1)^{2r}. \quad (17)$$

Consequently,

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} (\mathcal{L}_m(B) + 1) (C_0 \ln(p^a) + 1)^{2r}. \quad (18)$$

Proof. Fix a nonzero Walsh frequency $k \in \Delta_{2r}^*$. By Lemma 2,

$$N^*S(w_k, \mathcal{P}) = \sum_{u \in B_{\text{adm}}} \psi_1(\eta_k x(P_0 + [u]R) + \tilde{\eta}_k y(P_0 + [u]R)).$$

Define

$$f_{k, P_0}(P) := \eta_k x(P + P_0) + \tilde{\eta}_k y(P + P_0).$$

The polar divisor of f_{k, P_0} is supported at the single point $-P_0$, and $\deg(f_{k, P_0}) \leq 3$. As in the proof of Theorem 10, Lemma 5 shows that f_{k, P_0} is not of Artin–Schreier form. Applying Proposition 19 with $S = R$, $M = m$, and $A = B_{\text{adm}}$ gives

$$|N^*S(w_k, \mathcal{P})| \leq (1 + \deg(f_{k, P_0}))q^{1/2} \mathcal{L}_m(B_{\text{adm}}) \leq 4q^{1/2} \mathcal{L}_m(B_{\text{adm}}).$$

Applying Lemma 1 with $d = 2r$ yields (17). The rougher estimate (18) follows from Lemma 20. $\square$

4.4 Serial discrepancy in the general subperiod regime

Theorem 22 (General serial discrepancy bound via admissible Fourier mass). *Assume $|e| \geq 2$, $\gcd(e, p) = 1$, and $p \geq 5$. Fix $2 \leq s < t$ and let $\mathcal{P}^{(s)}$ be the admissible serial s -tuple collection, of cardinality N_s^* . Then*

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) \mathcal{L}_m(B_{s, \text{adm}}) (C_0 \ln(p^a) + 1)^{2rs}. \quad (19)$$

Consequently,

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\mathcal{L}_m(B) + s) (C_0 \ln(p^a) + 1)^{2rs}. \quad (20)$$

Proof. Each admissible serial block is parameterized by an index $u \in B_{s, \text{adm}}$ and has the form

$$(\Phi(A_0 + [u]R), \Phi(A_1 + [eu]R), \dots, \Phi(A_{s-1} + [e^{s-1}u]R)), \quad A_j := P_0 + [\beta_j]R.$$

Fix a nonzero Walsh frequency $k \in \Delta_{2rs}^*$. The corresponding Walsh coefficient equals a sum over $u \in B_{s, \text{adm}}$ of

$$\psi_1(g_k([u]R)),$$

where

$$g_k(P) := \sum_{j=0}^{s-1} \left(\eta_{k,j} x([e^j]P + A_j) + \tilde{\eta}_{k,j} y([e^j]P + A_j) \right) \in \mathbb{F}_q(E).$$

The proof of Theorem 11, based on Lemma 4 and Lemma 5, shows that

$$\deg(g_k) \leq 3 \sum_{j=0}^{s-1} e^{2j}$$

and that g_k is not of Artin–Schreier form. Applying Proposition 19 with $S = R$, $M = m$, and $A = B_{s, \text{adm}}$ yields

$$\left| N_s^* S(w_k, \mathcal{P}^{(s)}) \right| \leq 2 \deg(g_k) q^{1/2} \mathcal{L}_m(B_{s, \text{adm}}) \leq 6q^{1/2} \left(\sum_{j=0}^{s-1} e^{2j} \right) \mathcal{L}_m(B_{s, \text{adm}}).$$

Applying Lemma 1 in dimension $2rs$ yields (19). The rough bound (20) follows from Lemma 20. $\square$

4.5 Non-overlapping discrepancy in the general subperiod regime

Theorem 23 (General non-overlapping discrepancy bound via admissible Fourier mass). *Assume $|e| \geq 2$, $\gcd(e, p) = 1$, and $p \geq 5$. Fix $s \geq 2$ and let $\tilde{\mathcal{P}}^{(s)}$ be the admissible non-overlapping s -tuple collection, of cardinality $\tilde{N}_s^*$. Then*

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\tilde{N}_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) \mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) (C_0 \ln(p^a) + 1)^{2rs}. \quad (21)$$

Proof. For $u \in C_s$ and $0 \leq j \leq s-1$, one has

$$P_{ns+j} = T^j(P_{ns}) = P_j + [e^j u]R_s, \quad P_j := T^j(P_0).$$

Hence the admissible non-overlapping s -blocks are parameterized by $\tilde{B}_{s,\text{adm}}$ via

$$u \mapsto (\Phi(P_0 + [u]R_s), \Phi(P_1 + [eu]R_s), \dots, \Phi(P_{s-1} + [e^{s-1}u]R_s)).$$

Fix a nonzero Walsh frequency $k \in \Delta_{2rs}^*$. The corresponding Walsh coefficient equals a sum over $u \in \tilde{B}_{s,\text{adm}}$ of

$$\psi_1(\tilde{g}_k([u]R_s)),$$

where

$$\tilde{g}_k(P) := \sum_{j=0}^{s-1} \left(\eta_{k,j} x([e^j]P + P_j) + \tilde{\eta}_{k,j} y([e^j]P + P_j) \right) \in \mathbb{F}_q(E).$$

Exactly as in the proof of Theorem 11, using Lemma 4 and Lemma 5, one has

$$\deg(\tilde{g}_k) \leq 3 \sum_{j=0}^{s-1} e^{2j},$$

and $\tilde{g}_k$ is not of Artin-Schreier form. Applying Proposition 19 with $S = R_s$, $M = m_s$, and $A = \tilde{B}_{s,\text{adm}}$ yields

$$\left| \tilde{N}_s^* S(w_k, \tilde{\mathcal{P}}^{(s)}) \right| \leq 2 \deg(\tilde{g}_k) q^{1/2} \mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) \leq 6q^{1/2} \left(\sum_{j=0}^{s-1} e^{2j} \right) \mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}).$$

Applying Lemma 1 in dimension $2rs$ yields (21). □

Lemma 24 (Derived admissible Fourier mass). *One has*

$$\mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) \leq \mathcal{L}_{m_s}(C_s) + s.$$

Proof. Applying Lemma 7 to the purely periodic derived system

$$P_{(n+1)s} = [e^s]P_{ns} + Q_s$$

shows that

$$\gcd(e^s, m_s) = 1.$$

Hence for each $0 \leq j \leq s-1$, the map

$$u \mapsto e^j u \pmod{m_s}$$

is a bijection of $\mathbb{Z}/m_s\mathbb{Z}$. Therefore the condition

$$P_j + [e^j u]R_s = O$$

has at most one solution in $\mathbb{Z}/m_s\mathbb{Z}$, and so excludes at most one point of the derived index cycle C_s . Since $\tilde{B}_{s,\text{adm}}$ is obtained from C_s by imposing s such coordinate conditions, one has

$$|C_s \setminus \tilde{B}_{s,\text{adm}}| \leq s.$$

Lemma 20 at modulus m_s now gives

$$\mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) \leq \mathcal{L}_{m_s}(C_s) + s.$$

□

Corollary 25 (A rough non-overlapping Fourier-mass bound). *Under the hypotheses of Theorem 23,*

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\tilde{N}_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\mathcal{L}_{m_s}(C_s) + s) (C_0 \ln(p^a) + 1)^{2rs}. \quad (22)$$

Proof. Combine Theorem 23 with Lemma 24. □

4.6 Unconditional baseline bounds

Lemma 26 (Fourier ℓ^1 mass is bounded by the square root of the support size). *Let $M \geq 1$ and let $A \subset \mathbb{Z}/M\mathbb{Z}$. Then*

$$\mathcal{L}_M(A) \leq \sqrt{|A|}.$$

Proof. Parseval gives

$$\sum_{j=0}^{M-1} \left| \widehat{\mathbf{1}_A}^{(M)}(j) \right|^2 = M|A|.$$

Applying Cauchy–Schwarz,

$$\sum_{j=0}^{M-1} \left| \widehat{\mathbf{1}_A}^{(M)}(j) \right| \leq \sqrt{M} \sqrt{M|A|} = M \sqrt{|A|}.$$

Divide by M . □

Corollary 27 (Unconditional baseline for D). *Under the hypotheses of Theorem 21,*

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{\sqrt{N^*}} (C_0 \ln(p^a) + 1)^{2r}. \quad (23)$$

Proof. Apply Lemma 26 to $A = B_{\text{adm}} \subset \mathbb{Z}/m\mathbb{Z}$ and use $|A| = N^*$ in Theorem 21. □

Corollary 28 (Unconditional baseline for D_s). *Under the hypotheses of Theorem 22,*

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\sqrt{N_s^*}} \left(\sum_{j=0}^{s-1} e^{2j}\right) (C_0 \ln(p^a) + 1)^{2rs}. \quad (24)$$

Proof. Apply Lemma 26 to $A = B_{s,\text{adm}} \subset \mathbb{Z}/m\mathbb{Z}$ and use $|A| = N_s^*$ in Theorem 22. $\square$

Corollary 29 (Unconditional baseline for $\tilde{D}_s$). *Under the hypotheses of Theorem 23,*

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\sqrt{\tilde{N}_s^*}} \left(\sum_{j=0}^{s-1} e^{2j}\right) (C_0 \ln(p^a) + 1)^{2rs}. \quad (25)$$

Proof. Apply Lemma 26 to $A = \tilde{B}_{s,\text{adm}} \subset \mathbb{Z}/m_s\mathbb{Z}$ and use $|A| = \tilde{N}_s^*$ in Theorem 23. $\square$

5 Arithmetic structure of the index sequence

Theorem 21, Theorem 22, and Theorem 23 reduce the discrepancy problem to admissible Fourier masses of one-dimensional LCG index sets. This section isolates the arithmetic structure of those index sets.

5.1 Local behavior modulo prime powers

Write the prime-power factorization

$$m = \prod_{\ell} \ell^{\nu_{\ell}}.$$

For a prime $\ell \mid m$, let

$$a_{\ell} := v_{\ell}(e - 1).$$

Lemma 30 (Local behavior modulo prime powers). *Let $\ell^{\nu} \parallel m$ and set $a := v_{\ell}(e - 1)$. Then:*

(a) *For every $b \leq \min\{a, \nu\}$, one has*

$$\beta_n \equiv n \pmod{\ell^b}.$$

In particular, if $a \geq \nu$, then $\beta_n \equiv n \pmod{\ell^{\nu}}$ for all n .

(b) *If $a = 0$, then there exists a unique fixed point $\beta_{*,\ell} \in \mathbb{Z}/\ell^{\nu}\mathbb{Z}$ such that*

$$\beta_{*,\ell} \equiv e\beta_{*,\ell} + 1 \pmod{\ell^{\nu}},$$

and

$$\beta_n - \beta_{*,\ell} \equiv -\beta_{*,\ell} e^n \pmod{\ell^{\nu}}.$$

Proof. (a) If $e \equiv 1 \pmod{\ell^b}$ then the recursion (8) becomes

$$\beta_{n+1} \equiv \beta_n + 1 \pmod{\ell^b}, \quad \beta_0 \equiv 0,$$

hence $\beta_n \equiv n \pmod{\ell^b}$.

(b) If $a = 0$, then $1 - e$ is invertible modulo ℓ^{ν} , so the fixed-point congruence has a unique solution $\beta_{*,\ell}$. Subtracting the fixed-point relation from the recursion gives

$$(\beta_{n+1} - \beta_{*,\ell}) \equiv e(\beta_n - \beta_{*,\ell}) \pmod{\ell^{\nu}},$$

and iteration yields the stated formula. $\square$

Corollary 31 (No mixed local factors). *Assume that for every prime $\ell \mid m$, one has either $v_\ell(e - 1) = 0$ or $v_\ell(e - 1) \geq \nu_\ell$. Define*

$$m_{\text{tr}} := \prod_{v_\ell(e-1) \geq \nu_\ell} \ell^{\nu_\ell}, \quad m_{\text{pow}} := \prod_{v_\ell(e-1)=0} \ell^{\nu_\ell}, \quad m = m_{\text{tr}} m_{\text{pow}}.$$

Then $\gcd(m_{\text{tr}}, m_{\text{pow}}) = 1$, and there exists $\beta_ \in \mathbb{Z}/m_{\text{pow}}\mathbb{Z}$ such that under the Chinese remainder isomorphism*

$$\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times \mathbb{Z}/m_{\text{pow}}\mathbb{Z},$$

one has

$$\beta_n \mapsto (n \bmod m_{\text{tr}}, \beta_*(1 - e^n) \bmod m_{\text{pow}}).$$

Proof. For every prime power dividing m_{tr} , part (a) of Lemma 30 gives $\beta_n \equiv n$ modulo that prime power. For every prime power dividing m_{pow} , part (b) gives a local fixed point and hence

$$\beta_n \equiv \beta_{*,\ell}(1 - e^n) \pmod{\ell^{\nu_\ell}}.$$

Combining the local fixed points by the Chinese remainder theorem yields $\beta_* \in \mathbb{Z}/m_{\text{pow}}\mathbb{Z}$ with the claimed formula. $\square$

Remark 32. Lemma 30 shows that the unresolved arithmetic is concentrated in the *mixed* local prime powers, namely those with

$$0 < v_\ell(e - 1) < \nu_\ell.$$

If no such factors occur, the index sequence is explicitly described by Corollary 31 as a CRT combination of a translation component and a power-generator component.

Remark 33 (Conventions for trivial power components). If $m_{\text{pow}} = 1$, then the pure power component is trivial. In that case we set

$$\tau := 1, \quad G := \{0\} \subset \mathbb{Z}/1\mathbb{Z}.$$

Similarly, if $m_{s,\text{pow}} = 1$ in the derived setting, we set

$$\tau_s := 1, \quad G_s := \{0\} \subset \mathbb{Z}/1\mathbb{Z}.$$

With these conventions, all subsequent formulas remain valid without further case distinctions.

Theorem 34 (A strengthened reduction in the no-mixed regime). *Assume the hypotheses of Corollary 31. With the conventions of Remark 33, set*

$$\tau := \begin{cases} 1, & m_{\text{pow}} = 1, \\ \text{ord}_{m_{\text{pow}}}(e), & m_{\text{pow}} > 1, \end{cases} \quad G := \{\beta_*(1 - e^v) \bmod m_{\text{pow}} : 0 \leq v < \tau\} \subset \mathbb{Z}/m_{\text{pow}}\mathbb{Z}.$$

Assume in addition that

$$\gcd(m_{\text{tr}}, \tau) = 1.$$

Then the following hold.

(i) *The index period is*

$$t = m_{\text{tr}}\tau.$$

(ii) Under the Chinese remainder isomorphism

$$\mathbb{Z}/m\mathbb{Z} \cong \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times \mathbb{Z}/m_{\text{pow}}\mathbb{Z},$$

the index cycle

$$B = \{\beta_0, \dots, \beta_{t-1}\}$$

is exactly the Cartesian product

$$B = \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times G.$$

(iii) The normalized Fourier ℓ^1 mass satisfies

$$\mathcal{L}_m(B) = \mathcal{L}_{m_{\text{pow}}}(G).$$

Consequently,

$$\mathcal{L}_m(B_{\text{adm}}) \leq \mathcal{L}_{m_{\text{pow}}}(G) + 1, \quad \mathcal{L}_m(B_{s,\text{adm}}) \leq \mathcal{L}_{m_{\text{pow}}}(G) + s.$$

Proof. By Lemma 7, one has $\gcd(e, m) = 1$, hence in particular $\gcd(e, m_{\text{pow}}) = 1$, so $\tau = \text{ord}_{m_{\text{pow}}}(e)$ is well defined. By Corollary 31, under CRT one has

$$\beta_n \mapsto (n \bmod m_{\text{tr}}, \beta_*(1 - e^n) \bmod m_{\text{pow}}).$$

The first coordinate has period m_{tr} . The second coordinate has period τ : indeed, if

$$\beta_*(1 - e^{n+r}) \equiv \beta_*(1 - e^n) \pmod{m_{\text{pow}}},$$

then

$$\beta_* e^n (e^r - 1) \equiv 0 \pmod{m_{\text{pow}}}.$$

Since $(1-e)\beta_* \equiv 1 \pmod{m_{\text{pow}}}$, the element β_* is invertible modulo m_{pow} ; and since $\gcd(e, m_{\text{pow}}) = 1$, the element e^n is also invertible modulo m_{pow} . Therefore

$$e^r \equiv 1 \pmod{m_{\text{pow}}},$$

so $\tau \mid r$. Hence the second coordinate has exact period τ .

It follows that the full period of (β_n) modulo m is

$$t = \text{lcm}(m_{\text{tr}}, \tau) = m_{\text{tr}}\tau,$$

because $\gcd(m_{\text{tr}}, \tau) = 1$. This proves (i).

To prove (ii), let $(u, v) \in \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times \{0, \dots, \tau - 1\}$. By the Chinese remainder theorem there exists a unique residue class $n \bmod t$ such that

$$n \equiv u \pmod{m_{\text{tr}}}, \quad n \equiv v \pmod{\tau}.$$

Since $e^n \equiv e^v \pmod{m_{\text{pow}}}$, Corollary 31 gives

$$\beta_n \mapsto (u, \beta_*(1 - e^v)).$$

Thus every element of $\mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times G$ occurs in B . Conversely, every β_n clearly has this form, proving

$$B = \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times G.$$

For (iii), observe first that the normalized Fourier ℓ^1 mass is invariant under additive-group isomorphisms, since such an isomorphism identifies the character groups and preserves the normalized ℓ^1 norm of the Fourier transform. Hence one may compute $\mathcal{L}_m(B)$ on the product group

$$\mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times \mathbb{Z}/m_{\text{pow}}\mathbb{Z}$$

using product characters

$$\chi_{a,b}(x, y) := \exp\left(-2\pi i \frac{ax}{m_{\text{tr}}}\right) \exp\left(-2\pi i \frac{by}{m_{\text{pow}}}\right),$$

with $0 \leq a < m_{\text{tr}}$ and $0 \leq b < m_{\text{pow}}$. Since $B = \mathbb{Z}/m_{\text{tr}}\mathbb{Z} \times G$, one has

$$\widehat{\mathbf{1}}_B(a, b) = \sum_{x \in \mathbb{Z}/m_{\text{tr}}\mathbb{Z}} e^{-2\pi i ax/m_{\text{tr}}} \sum_{y \in G} e^{-2\pi i by/m_{\text{pow}}}.$$

The first factor vanishes unless $a = 0$, in which case it equals m_{tr} . Therefore

$$\sum_{a=0}^{m_{\text{tr}}-1} \sum_{b=0}^{m_{\text{pow}}-1} |\widehat{\mathbf{1}}_B(a, b)| = m_{\text{tr}} \sum_{b=0}^{m_{\text{pow}}-1} |\widehat{\mathbf{1}}_G^{(m_{\text{pow}})}(b)|.$$

Dividing by $m = m_{\text{tr}}m_{\text{pow}}$ yields

$$\mathcal{L}_m(B) = \mathcal{L}_{m_{\text{pow}}}(G).$$

The final inequalities now follow from Lemma 20. $\square$

Corollary 35 (Improved discrepancy bounds in the no-mixed regime). *Under the hypotheses of Theorem 34,*

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} (\mathcal{L}_{m_{\text{pow}}}(G) + 1) (C_0 \ln(p^a) + 1)^{2r}, \quad (26)$$

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\mathcal{L}_{m_{\text{pow}}}(G) + s) (C_0 \ln(p^a) + 1)^{2rs}. \quad (27)$$

In particular, since $\mathcal{L}_{m_{\text{pow}}}(G) \leq \sqrt{\tau}$,

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} (\sqrt{\tau} + 1) (C_0 \ln(p^a) + 1)^{2r}, \quad (28)$$

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\sqrt{\tau} + s) (C_0 \ln(p^a) + 1)^{2rs}. \quad (29)$$

Proof. Combine Theorem 34 with Theorem 21, Theorem 22, and Lemma 26. $\square$

Theorem 36 (A strengthened reduction for the derived cycle in the no-mixed regime). *Fix $s \geq 2$ and consider the derived system*

$$P_{(n+1)s} = [e^s]P_{ns} + Q_s, \quad Q_s := \sum_{j=0}^{s-1} [e^j]Q, \quad R_s := P_s - P_0.$$

Let

$$m_s := \text{ord}(R_s), \quad C_s := \{\gamma_0, \dots, \gamma_{L_s-1}\} \subset \mathbb{Z}/m_s\mathbb{Z},$$

where

$$\gamma_{n+1} \equiv e^s \gamma_n + 1 \pmod{m_s}, \quad \gamma_0 \equiv 0, \quad L_s = \frac{t}{\gcd(t, s)}.$$

Assume that for every prime $\ell \mid m_s$, one has either $v_\ell(e^s - 1) = 0$ or $v_\ell(e^s - 1) \geq \nu_{\ell, s}$, where

$$m_s = \prod_{\ell} \ell^{\nu_{\ell, s}}.$$

Define

$$m_{s, \text{tr}} := \prod_{v_\ell(e^s - 1) \geq \nu_{\ell, s}} \ell^{\nu_{\ell, s}}, \quad m_{s, \text{pow}} := \prod_{v_\ell(e^s - 1) = 0} \ell^{\nu_{\ell, s}}, \quad m_s = m_{s, \text{tr}} m_{s, \text{pow}},$$

and, with the conventions of Remark 33, set

$$\tau_s := \begin{cases} 1, & m_{s, \text{pow}} = 1, \\ \text{ord}_{m_{s, \text{pow}}}(e^s), & m_{s, \text{pow}} > 1. \end{cases}$$

Assume moreover that

$$\gcd(m_{s, \text{tr}}, \tau_s) = 1.$$

Then the following hold.

(i) The derived index period is

$$L_s = m_{s, \text{tr}} \tau_s.$$

(ii) There exists $\gamma_{*, s} \in \mathbb{Z}/m_{s, \text{pow}}\mathbb{Z}$ such that, under the Chinese remainder isomorphism

$$\mathbb{Z}/m_s\mathbb{Z} \cong \mathbb{Z}/m_{s, \text{tr}}\mathbb{Z} \times \mathbb{Z}/m_{s, \text{pow}}\mathbb{Z},$$

the derived cycle is exactly

$$C_s = \mathbb{Z}/m_{s, \text{tr}}\mathbb{Z} \times G_s, \quad G_s := \{\gamma_{*, s}(1 - (e^s)^v) \bmod m_{s, \text{pow}} : 0 \leq v < \tau_s\}.$$

(iii) The normalized Fourier ℓ^1 mass satisfies

$$\mathcal{L}_{m_s}(C_s) = \mathcal{L}_{m_{s, \text{pow}}}(G_s).$$

Consequently,

$$\mathcal{L}_{m_s}(\tilde{B}_{s, \text{adm}}) \leq \mathcal{L}_{m_{s, \text{pow}}}(G_s) + s.$$

Proof. If $m_{s, \text{pow}} = 1$, then $\tau_s = 1$ and $G_s = \{0\}$ by Remark 33. The statements are immediate, so assume $m_{s, \text{pow}} > 1$. Apply Lemma 7 to the purely periodic derived system to obtain

$$\gcd(e^s, m_s) = 1.$$

Therefore the local analysis of Lemma 30 and Corollary 31 applies to the derived LCG

$$\gamma_{n+1} \equiv e^s \gamma_n + 1 \pmod{m_s}$$

with multiplier e^s and modulus m_s . This yields a decomposition

$$\gamma_n \mapsto (n \bmod m_{s,\text{tr}}, \gamma_{*,s}(1 - (e^s)^n) \bmod m_{s,\text{pow}})$$

under CRT, where $\gamma_{*,s}$ is the fixed point modulo $m_{s,\text{pow}}$. Exactly as in the proof of Theorem 34, the first coordinate has period $m_{s,\text{tr}}$, the second has exact period τ_s , and the coprimality hypothesis implies

$$L_s = \text{lcm}(m_{s,\text{tr}}, \tau_s) = m_{s,\text{tr}}\tau_s.$$

This proves (i).

The same CRT argument as in Theorem 34 shows that every pair

$$(u, v) \in \mathbb{Z}/m_{s,\text{tr}}\mathbb{Z} \times \{0, \dots, \tau_s - 1\}$$

occurs uniquely in the derived cycle, proving

$$C_s = \mathbb{Z}/m_{s,\text{tr}}\mathbb{Z} \times G_s.$$

This proves (ii).

For (iii), compute the Fourier transform of $\mathbf{1}_{C_s}$ on the product group

$$\mathbb{Z}/m_{s,\text{tr}}\mathbb{Z} \times \mathbb{Z}/m_{s,\text{pow}}\mathbb{Z}.$$

As in the proof of Theorem 34, the full translation factor forces vanishing of all nontrivial frequencies in the first coordinate, and one obtains

$$\mathcal{L}_{m_s}(C_s) = \mathcal{L}_{m_{s,\text{pow}}}(G_s).$$

Finally, Lemma 24 gives

$$\mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) \leq \mathcal{L}_{m_s}(C_s) + s = \mathcal{L}_{m_{s,\text{pow}}}(G_s) + s.$$

□

Corollary 37 (Improved non-overlapping discrepancy bounds in the derived no-mixed regime). *Under the hypotheses of Theorem 36,*

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\tilde{N}_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\mathcal{L}_{m_{s,\text{pow}}}(G_s) + s) (C_0 \ln(p^a) + 1)^{2rs}, \quad (30)$$

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\tilde{N}_s^*} \left(\sum_{j=0}^{s-1} e^{2j}\right) (\sqrt{\tau_s} + s) (C_0 \ln(p^a) + 1)^{2rs}. \quad (31)$$

Proof. Combine Theorem 36 with Corollary 25 and Lemma 26. □

Theorem 38 (Pure power representation of the Fourier masses in the no-mixed regime). *Under the hypotheses of Theorem 34, one has*

$$\mathcal{L}_{m_{\text{pow}}}(G) = \frac{1}{m_{\text{pow}}} \sum_{b=0}^{m_{\text{pow}}-1} \left| \sum_{v=0}^{\tau-1} \exp\left(\frac{2\pi i b e^v}{m_{\text{pow}}}\right) \right|. \quad (32)$$

Under the hypotheses of Theorem 36, one has

$$\mathcal{L}_{m_{s,\text{pow}}}(G_s) = \frac{1}{m_{s,\text{pow}}} \sum_{b=0}^{m_{s,\text{pow}}-1} \left| \sum_{v=0}^{\tau_s-1} \exp\left(\frac{2\pi i b (e^s)^v}{m_{s,\text{pow}}}\right) \right|. \quad (33)$$

Proof. We prove (32); the proof of (33) is identical with e replaced by e^s . By Theorem 34,

$$G = \{\beta_*(1 - e^v) \bmod m_{\text{pow}} : 0 \leq v < \tau\}.$$

Since $(1 - e)\beta_* \equiv 1 \pmod{m_{\text{pow}}}$, the element β_* is invertible modulo m_{pow} . Therefore, for each b modulo m_{pow} ,

$$\sum_{g \in G} \exp\left(-\frac{2\pi i b g}{m_{\text{pow}}}\right) = \exp\left(-\frac{2\pi i b \beta_*}{m_{\text{pow}}}\right) \sum_{v=0}^{\tau-1} \exp\left(\frac{2\pi i b \beta_* e^v}{m_{\text{pow}}}\right).$$

Taking absolute values removes the leading phase. Because multiplication by the unit β_* permutes the residue classes modulo m_{pow} , averaging over b gives

$$\mathcal{L}_{m_{\text{pow}}}(G) = \frac{1}{m_{\text{pow}}} \sum_{b=0}^{m_{\text{pow}}-1} \left| \sum_{v=0}^{\tau-1} \exp\left(\frac{2\pi i b e^v}{m_{\text{pow}}}\right) \right|.$$

□

Corollary 39 (Transfer principle from exponential sums along multiplicative orbits). *Assume the hypotheses of Theorem 34 and $m_{\text{pow}} > 1$. Suppose that for some quantity $\mathcal{E}(m_{\text{pow}}, \tau, e)$ one has*

$$\max_{1 \leq b < m_{\text{pow}}} \left| \sum_{v=0}^{\tau-1} \exp\left(\frac{2\pi i b e^v}{m_{\text{pow}}}\right) \right| \leq \mathcal{E}(m_{\text{pow}}, \tau, e).$$

Then

$$\mathcal{L}_{m_{\text{pow}}}(G) \leq \frac{\tau + (m_{\text{pow}} - 1)\mathcal{E}(m_{\text{pow}}, \tau, e)}{m_{\text{pow}}}. \quad (34)$$

Consequently,

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} \left(\frac{\tau + (m_{\text{pow}} - 1)\mathcal{E}(m_{\text{pow}}, \tau, e)}{m_{\text{pow}}} + 1 \right) (C_0 \ln(p^a) + 1)^{2r}, \quad (35)$$

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j} \right) \left(\frac{\tau + (m_{\text{pow}} - 1)\mathcal{E}(m_{\text{pow}}, \tau, e)}{m_{\text{pow}}} + s \right) (C_0 \ln(p^a) + 1)^{2rs}. \quad (36)$$

Similarly, if $m_{s,\text{pow}} > 1$ and under the hypotheses of Theorem 36, any estimate

$$\max_{1 \leq b < m_{s,\text{pow}}} \left| \sum_{v=0}^{\tau_s-1} \exp\left(\frac{2\pi i b (e^s)^v}{m_{s,\text{pow}}}\right) \right| \leq \mathcal{E}_s(m_{s,\text{pow}}, \tau_s, e^s)$$

implies

$$D(\tilde{\mathcal{P}}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{\tilde{N}_s^*} \left(\sum_{j=0}^{s-1} e^{2j} \right) \left(\frac{\tau_s + (m_{s,\text{pow}} - 1)\mathcal{E}_s(m_{s,\text{pow}}, \tau_s, e^s)}{m_{s,\text{pow}}} + s \right) (C_0 \ln(p^a) + 1)^{2rs}. \quad (37)$$

Proof. The bound (34) follows immediately from Theorem 38 by isolating the term $b = 0$, which contributes τ , and bounding the remaining $m_{\text{pow}} - 1$ terms by $\mathcal{E}(m_{\text{pow}}, \tau, e)$. The discrepancy bounds then follow from Corollary 35 and Corollary 37. □

Proposition 40 (Prime-modulus special case). *Assume the hypotheses of Theorem 34, and suppose in addition that $m_{\text{pow}} = \ell$ is prime. Then*

$$\mathcal{L}_\ell(G) \leq \frac{\tau + (\ell - 1)\ell^{1/2}}{\ell}.$$

Consequently,

$$D(\mathcal{P}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2r} + \frac{4q^{1/2}}{N^*} \left(\frac{\tau + (\ell - 1)\ell^{1/2}}{\ell} + 1 \right) (C_0 \ln(p^a) + 1)^{2r}, \quad (38)$$

$$D(\mathcal{P}^{(s)}) \leq 1 - \left(1 - \frac{1}{p^a}\right)^{2rs} + \frac{6q^{1/2}}{N_s^*} \left(\sum_{j=0}^{s-1} e^{2j} \right) \left(\frac{\tau + (\ell - 1)\ell^{1/2}}{\ell} + s \right) (C_0 \ln(p^a) + 1)^{2rs}. \quad (39)$$

Proof. Since ℓ is prime and $\tau = \text{ord}_\ell(e)$, the set

$$\{e^v \bmod \ell : 0 \leq v < \tau\}$$

is the multiplicative subgroup $H = \langle e \rangle \subset \mathbb{F}_\ell^\times$ of size τ . For $b \not\equiv 0 \pmod{\ell}$ one has

$$\sum_{v=0}^{\tau-1} \exp\left(\frac{2\pi i b e^v}{\ell}\right) = \sum_{x \in H} e_\ell(bx), \quad e_\ell(z) := \exp(2\pi i z / \ell).$$

Let $d = (\ell - 1)/\tau$. The indicator of H in $\mathbb{F}_\ell^\times$ is

$$\mathbf{1}_H(x) = \frac{1}{d} \sum_{\chi^d=1} \chi(x),$$

where the sum ranges over the d multiplicative characters of $\mathbb{F}_\ell^\times$ of order dividing d . Hence

$$\sum_{x \in H} e_\ell(bx) = \frac{1}{d} \sum_{\chi^d=1} \sum_{x \in \mathbb{F}_\ell^\times} \chi(x) e_\ell(bx).$$

For the trivial character the inner sum equals -1 , while for every nontrivial character it is a Gauss sum of absolute value $\ell^{1/2}$. Therefore

$$\left| \sum_{x \in H} e_\ell(bx) \right| \leq \frac{1 + (d-1)\ell^{1/2}}{d} \leq \ell^{1/2} \quad (b \not\equiv 0 \pmod{\ell}).$$

The term $b = 0$ contributes τ . Applying Corollary 39 with $\mathcal{E}(\ell, \tau, e) = \ell^{1/2}$ yields the result. $\square$

Remark 41. Theorem 38 shows that, in the no-mixed regimes, the remaining analytic input is an estimate for additive exponential sums along the multiplicative orbit

$$1, e, e^2, \dots, e^{\tau-1} \pmod{m_{\text{pow}}},$$

and similarly for the derived orbit generated by e^s modulo $m_{s, \text{pow}}$. Accordingly, any explicit bound for such exponential sums can be substituted into Corollary 39.

5.2 Analytic consequences

The discrepancy bounds in Section 4 show that any improvement of the form

$$\mathcal{L}_m(B_{\text{adm}}) \ll (\log m)^A, \quad \mathcal{L}_m(B_{s,\text{adm}}) \ll (\log m)^A, \quad \mathcal{L}_{m_s}(\tilde{B}_{s,\text{adm}}) \ll (\log m_s)^A$$

would immediately produce near Mok-type $q^{1/2}/t$ bounds up to polylogarithmic losses. This suggests three concrete directions. In the no-mixed regime, however, the present paper already reduces the problem completely to the pure power components G and G_s :

1. interval estimates in the pure translation local regime;
2. exponential-sum estimates for multiplicative orbits in the pure power local regime; in the no-mixed regime, Theorem 34, Theorem 36, and Theorem 38 reduce the relevant Fourier masses to explicit averages of such sums for the pure power components G and G_s ;
3. bilinear-sum methods over elliptic curves, following Shparlinski [4] and Ahmadi–Shparlinski [5], to treat the mixed regime directly.

6 Summary of results

The principal results are as follows.

1. In the full-coset regime (RMPC), one proves Mok-type $q^{1/2}/t$ bounds for the extreme discrepancy D , the serial discrepancy D_s , and, under the derived RMPC, the non-overlapping discrepancy $\tilde{D}_s$.
2. In the general subperiod regime, one reduces the full problem for D , D_s , and $\tilde{D}_s$ to admissible Fourier ℓ^1 masses of one-dimensional LCG index sets, and deduces unconditional baseline bounds of order $q^{1/2}/\sqrt{N^*}$, $q^{1/2}/\sqrt{N_s^*}$, and $q^{1/2}/\sqrt{\tilde{N}_s^*}$. In the no-mixed regime one further reduces these Fourier masses to the pure power components G and G_s , and then rewrites them exactly as averages of additive exponential sums along multiplicative orbits.
3. The arithmetic obstruction to stronger bounds is isolated in the mixed local prime-power regime $0 < v_\ell(e-1) < v_\ell$. In the complementary no-mixed regime, Theorem 34, Theorem 36, and their corollaries yield genuine strengthenings of the general bounds for D , D_s , and $\tilde{D}_s$.

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